INCH-POUND
MIL-DTL-2729E(SH)
25 June 2026
SUPERSEDING
MIL-R-2729D(SH)
28 January 1992

DETAIL SPECIFICATION

REGULATOR-EXCITER SYSTEMS, VOLTAGE, AC GENERATOR, NAVAL SHIPBOARD USE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high impact (HI) shockproof alternating current (AC) generator voltage regulator-exciter systems for use on naval surface ships and submarines.

1.2 Classification. Regulator-exciter systems are of the following types, as specified (see 6.2):

a. Type I – not used.

b. Type II – rotary brushless exciter and voltage regulator system (see 6.4.i).

c. Type III – combined static exciter and voltage regulator system (see 6.4.j).

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3 and 4 of this specification. This section does not include publications cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of publications cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-917	-	Electric Power Equipment, Basic Requirements for
MIL-I-1361	-	Instrument Auxiliaries, Electrical Measuring: Shunts, Resistors, and Transformers
MIL-DTL-2036	-	Enclosures for Electric and Electronic Equipment, Naval Shipboard

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using ASSIST at https://assist.dla.mil.

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MIL-DTL-2212	-	Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices
MIL-G-3087	-	Generator Sets, Steam Turbine (Direct and Alternating Current) (Naval Shipboard Use)
MIL-DTL-3124	-	Generator, Alternating Current, 60-Hertz (Naval Shipboard Use)
MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-P-15024/5	-	Plate, Identification
MIL-PRF-15160	-	Fuses, Instrument, Power, and Telephone General Specification for
MIL-DTL-15291	-	Switches, Rotary, Snap Action and Detent/Spring Return Action, General Specification for
MIL-T-16315	-	Transformers, Power, Step-Down (Miscellaneous, Naval Shipboard Use)
MIL-M-19097	-	Motor-Generators, DC to AC, Shipboard Service NOTE: MIL-M-19097 is inactive for new design and intended only for guidance.
MIL-PRF-19500	-	Semiconductor Devices, General Specification for
MIL-G-21296	-	Generator Sets, Diesel Engine Direct and Alternating Current (Naval Shipboard Use)
MIL-G-22077	-	Generator Sets, Gas Turbine, Direct and Alternating Current, Naval Shipboard Use
MIL-DTL-26482	-	Connectors, Electrical, Circular, Miniature, Quick Disconnect, Environment Resisting, Receptacles and Plugs, General Specification for
MIL-DTL-28748	-	Connector, Plug and Receptacle, Rectangular, Rack and Panel, Solder Type and Crimp Type Contacts, General Specification for
MIL-DTL-32613	-	Controller, Auxiliary System, Naval Shipboard Use
MIL-PRF-38535	-	Integrated Circuits (Microcircuits) Manufacturing, General Specification for
MIL-DTL-55302	-	Connectors, Printed Circuit Subassembly and Accessories
MIL-DTL-83723	-	Connectors, Electrical, Circular, Environment Resisting and Hermetically Sealed, Receptacles and Plugs, General Specification for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-1399-300-1	-	Department of Defense Interface Standard, Section 300, Part 1, Low Voltage Electric Power, Alternating Current

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-217	-	Reliability Prediction of Electronic Equipment
MIL-HDBK-338	-	Electronic Reliability Design Handbook
MIL-HDBK-454	-	General Guideline for Electronic Equipment
MIL-HDBK-470	-	Designing and Developing Maintainable Products and Systems, Volume 1
MIL-HDBK-472	-	Maintainability Prediction
MIL-HDBK-781	-	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production
MIL-HDBK-1331	-	Parameters to be Controlled for the Specification of Microcircuits, Handbook for

(Copies of these publications are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE (DOD) PUBLICATIONS

DoDI 8500.01 - Cybersecurity

(Copies of this publication are available online at www.esd.whs.mil/DD/.)

FEDERAL INFORMATION PROCESSING STANDARDS (FIPS) PUBLICATIONS

FIPS PUB 180-4 - Secure Hash Standard (SHS)

(Copies of this publication are available online at https://csrc.nist.rip/publications/PubsFIPS.html.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SPECIAL PUBLICATION 800-63 - Digital Identity Guidelines

(Copies of this publication are available online at http://nvl.nist.gov/.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9086-KC-STM-010/300	-	NSTM Chapter 300, Electric Plant - General
T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process
0967-LP-312-8010	-	Designers of Shipboard Electronic Equipment, Maintainability Design Criteria Handbook

(Copies of these publications are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required). These publications are available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

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2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this publication are available online at www.aia-aerospace.org.)

SAE INTERNATIONAL

SAE AS50151 - Connectors, Electrical, Circular Threaded, AN Type, General Specification for

SAE JA1000/1 - Reliability Program Standard Implementation Guide

(Copies of these publications are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. Materials shall be in accordance with MIL-DTL-917.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Nonstructural parts. Nonstructural parts such as terminal boxes, housing, ducts, and similar parts shall be fabricated from aluminum alloy, where feasible.

3.2.3 Toxicity. When evaluated in accordance with 4.6, the voltage regulator exciter system shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.6 and 6.9).

3.2.4 Prohibited materials. The voltage regulator-exciter system shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

3.3 Design.

3.3.1 General. Unless otherwise specified herein, regulators and exciters shall conform to MIL-DTL-917.

3.3.2 Reliability and maintainability.

3.3.2.1 Interchangeability. As specified in MIL-DTL-917, in no case shall the mechanical parts of the equipment be physically interchangeable or reversible unless such parts are also interchangeable or reversible with regard to function, performance, and strength. The voltage regulator and the exciter shall each be evaluated independently to ensure compliance to this requirement.

3.3.2.2 Definitions. Unless otherwise specified herein, MIL-HDBK-338 should be used as a guide for definitions of terms for reliability and maintainability.

3.3.2.3 Reliability. The reliability measure is mean time between failure (MTBF) as defined in MIL-HDBK-781. The MTBF shall be greater than 25,000 hours.

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3.3.2.3.1 Reliability prediction. A reliability prediction analysis should be done using a parts stress analysis. MIL-HDBK-217 should be used as a guide. As specified (see 6.2), the reliability predictions report shall be provided.

3.3.2.3.2 Equipment failure. Voltage regulator-exciter system failure shall be defined as inability of the system to perform and function within the requirements of this specification.

3.3.2.3.3 Stress levels. Stress levels shall be calculated from design data using MIL-HDBK-217 as a guide.

3.3.2.4 Maintainability.

3.3.2.4.1 Maintainability measure. The measure of maintainability shall be equipment repair time (ERT) using MIL-HDBK-470 as a guide. The ERT shall not be greater than 3 hours ($ERT_{SPECIFIED}$). The maximum value of ERT that should be accepted no more than 10 percent of the time shall be no greater than 10 hours (ERT_{MAX}).

3.3.2.4.2 Maintainability design. Maintainability design shall be in accordance with NAVSEA 0967-LP-312-8010. The design should incorporate methods to protect personnel from inadvertent contact with voltages capable of producing shock hazards. MIL-HDBK-454 should be used as a guide. The voltage regulator layout shall provide adequate clearance and test points for use of standard multimeter test leads. Test points shall be provided where practicable to simplify routine maintenance, troubleshooting, and calibration of the equipment (for example, checking voltages, currents, and waveforms and the insertion of test signals). The test points shall be capable of being utilized during normal operation without affecting the equipment performance. The layout shall not require removal of multiple lowest replaceable units (LRU) to replace a single LRU unless otherwise approved by the buyer. When specified (see 6.2), the use of touchscreen digital interfaces shall be readily accessible at the front panel for use, or potentially exposed energized circuits shall be adequately covered or safely isolated from the technician such that inadvertent contact is a low risk while utilizing the touchscreen digital interface.

3.3.2.4.3 Maintainability prediction. Maintainability prediction shall be done using MIL-HDBK-472, procedure II, as a guide. A maintainability predictions report shall be submitted, if required (see 6.2).

3.3.2.5 Failure modes, effects, and criticality analysis. The failure modes, effects, and criticality analysis (FMECA) shall be in accordance with SAE JA1000/1, appendix A.10, and a report shall be submitted as specified (see 6.2). Critical items shall be identified.

3.3.3 Detail requirements.

3.3.3.1 Basic requirements. The procurement activity shall specify requirements for system drawings/documentation (see 6.2). Exciter system insulation shall be class F, or a greater temperature class as selected by the manufacturer in accordance with MIL-DTL-917. The voltage regulator-exciter system shall control the generator terminal voltage during all modes of operations and provide for stable operation when paralleled with other AC power sources. The major functions of the voltage regulator-exciter system are to provide automatic voltage regulation, provide overvoltage limiting (OVL), provide over excitation protection, and provide fault detection indications for critical fault conditions that prevent the voltage regulator from controlling generator field and non-critical fault conditions that do not prevent the voltage regulator from controlling the generator field.

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3.3.3.1.1 System description. The voltage regulator-exciter system shall consist of dual, redundant channels (see 6.4.c) (channel A or channel B), automatic voltage regulator, and associated circuitry. Each channel (channel A or channel B) may be the active (see 6.4.a) or inactive (see 6.4.e) channel. Each channel is one of two redundant halves of the voltage regulator system. The channel is said to be active if it has been selected to control the generator field; only one channel can be active at any time. The inactive channel is not allowed to control the generator field. Each channel (channel A or channel B) may also be considered available (see 6.4.b) or unavailable (see 6.4.k). A given channel of the system is said to be available to control the generator field if, at a minimum, power is applied to that channel and no faults that prevent the voltage regulator from controlling the generator field are present on that channel. A given channel of the system is said to be unavailable to control the generator field if, at a minimum, no power is applied to the channel or faults that prevent the voltage regulator from controlling the generator field are present. If a rotary brushless exciter is used, it shall conform to the construction and design requirements of MIL-DTL-3124. Semiconductors used in the rotating rectifier assembly shall be as specified in 3.3.3.5. All verification testing specified in MIL-DTL-3124 shall be performed with the rotary brushless exciter supplying excitation to the field of the main generator. Requirements for test procedures and reports shall be as specified (see 6.2).

3.3.3.1.2 Operational performance characteristics. If both channels are powered up simultaneously, either of the single channels with no known faults may assume control (either channel A or channel B). An automatic transfer of control occurs when there is a failure in the active channel. Manual transfers of control may be initiated by the operator. Both channels receive sensing signals. Only the active channel initiates over voltage limiting and over excitation protection. If both channels (channel A and channel B) experience critical fault conditions that prevent the voltage regulator from controlling generator field, the voltage regulator-exciter system shall deenergize to protect the generator. A means shall be provided to galvanically and electrically isolate each redundant half (channel A or channel B) of the voltage regulator system from all power sources so that maintenance can be performed on the inactive channel without shutting down the generator.

3.3.3.2 Enclosure and mounting.

3.3.3.2.1 Type II system. Enclosure and mounting shall be as follows:

a. Rotary exciter – enclosure and mounting shall be as specified (see 6.2).

b. Voltage regulator – protected, dedicated enclosure for deck mounting.

c. Front panel operating controllers and indicators – shall be in accordance with MIL-DTL-917.

3.3.3.2.2 Type III system. The enclosure shall be either arrangement A or arrangement B as specified (see 6.2).

3.3.3.2.2.1 Arrangement A. Enclosures and mounting shall be as follows:

a. Static exciter including field rectifiers – protected enclosure for deck-mounting.

b. Voltage regulator – protected enclosure for deck-mounting.

c. Front panel operating controllers and indicators – shall be in accordance with MIL-DTL-917.

3.3.3.2.2.2 Arrangement B. Enclosure and mounting shall be the same as arrangement A, except that the regulator shall be mounted in the exciter enclosure or dedicated enclosure in accordance with 3.3.3.2.4.

3.3.3.2.3 Ambient. Deck-mounted equipment shall be designed for a 50 degree Celsius (°C) ambient. Ambient temperature ranges shall be in accordance with MIL-DTL-917. The equipment shall have an operating temperature range of 0 to 50 °C. The equipment shall have a non-operating temperature range of -40 to 75 °C.

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3.3.3.2.4 Enclosures. Enclosures shall be constructed of corrosion-resisting material or material treated to resist corrosion in accordance with MIL-DTL-917. Enclosures shall be spray-tight protected in accordance with MIL-DTL-2036. Ready accessibility of electrical connections, test points, and adjustments shall be provided for convenience in installation and maintenance. Wherever possible, line voltage connections and control circuit test points shall be separated to permit safe troubleshooting with equipment operating. Enclosures and framework shall be constructed of steel, aluminum, or both. The use of aluminum for weight reduction is considered desirable and shall be used if structurally suitable. Enclosures shall be of the dead-front type.

3.3.3.2.5 Cooling. Unless otherwise specified (see 6.2), all components not mounted in the generator airstream shall be cooled by natural convection. Unless otherwise specified (see 6.2) or stated in MIL-DTL-3124, only those components that require forced air-cooling shall be mounted in the airstream. No sensitive equipment shall be mounted in the airstream.

3.3.3.2.6 Creepage and clearance. Unless otherwise specified (see 6.2), all electrical equipment shall be designed with creepage and clearance distances as specified in MIL-DTL-917, except the allowable electrical creepage for “up to 64 Volts AC or DC”, “set C” shall be changed to ½ inch for “Open” enclosures and ⅜ inch for “Enclosed” enclosures. For insulated bus, the insulation system is considered secondary to air and should not be used to reduce the creepage and clearance distance between bus conductors of opposite phase or polarity or between phase and ground.

3.3.3.3 Resistors, capacitors, and rheostats. Resistors, capacitors, and rheostats (if used) shall be selected and applied in accordance with MIL-DTL-917.

3.3.3.4 Transformer application. Separate control power, current, potential, and combination transformers shall be used with each regulator-exciter system, and no other burden shall be placed on the transformers. Transformers shall be ungrounded.

3.3.3.4.1 Power transformers. Control power transformers shall be in accordance with type SA of MIL-T-16315 and shall conform to the materials and methods of manufacture of MIL-DTL-917, except that allowable temperature limits shall be as specified in table I.

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TABLE I. Maximum permissible temperature (°C) – (continuous rated load).

Name of part [1/]	Insulation class		
	F	H	N
Transformer coils:			
Measured by the "resistance" method of MIL-DTL-917	125	150	170
Cores and mechanical parts adjacent to insulation – measured by the "thermometer" method of MIL-DTL-917	125	150	170
	All classes		
Capacitors – air within ½ inch of case:	85		
Semiconductor devices:	[2/]		
Resistors:			
Bare resistor material	425		
Resistor embedding material	350		
Bolted connection and terminal studs:			
Not plated	100		
Silver plated	115		
Contact:			
Solid	120		
Solid silver faced	130		

NOTES:

1/ Where no measurement method is indicated, thermometer or thermocouple method shall be used.

2/ Silicon rectifiers, voltage reference elements, voltage regulator elements, and transistors: permissible temperature rise on these parts is dependent upon operating condition. Maximum permissible case temperature shall be shown on the equipment drawing.

3.3.3.4.2 Current and potential transformers. Current, potential, or combination transformers shall be in accordance with MIL-I-1361 and shall conform to the materials and methods of manufacture specified in MIL-DTL-917, except that the allowable temperatures shall be as specified in table I.

3.3.3.4.3 Inductors. Inductors shall be in accordance with the materials and methods of manufacture specified in MIL-DTL-917, except the allowable temperature limits shall be as specified in table I.

3.3.3.5 Semiconductor devices. Semiconductors, such as diodes and controlled rectifiers used in the rotating rectifier assembly, shall be selected and applied in accordance with MIL-DTL-917. They shall be of the silicon type and shall be replaceable at the LRU.

3.3.3.5.1 Microcircuits. All integrated circuits shall conform to the requirements of MIL-PRF-38535, class B. MIL-HDBK-1331 should be used as a guide for selecting integrated circuits.

3.3.3.5.2 Discrete devices. All semiconductor devices, other than those specified in 3.3.3.5.1, shall conform to the requirements of MIL-PRF-19500 and MIL-DTL-917.

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3.3.3.6 Protective devices. When specified (see 6.2), protective devices, such as fuses, shall be provided in generator terminal voltage sensing, line current sensing, and field excitation circuits. If protective devices are used in the sensing or excitation circuits, the underlying control and protection system shall account for a loss of these circuits and loss of sensing protection is required (see 3.4.10.4). Unless otherwise specified (see 6.2), fuses shall be in accordance with MIL-PRF-15160. Fuses and fuse holders shall be selected and applied in accordance with materials and methods of manufacture specified in MIL-DTL-917.

3.3.3.7 Operating modes. The regulator-exciter system shall have the following operating modes:

a. Normal. In "Normal" mode, the voltage regulator-exciter system selects the active channel (either channel A or channel B) depending on availability of each channel. The active channel shall monitor the generator terminal voltage and adjust the exciter field excitation to regulate voltage. If a failure or loss of power occurs to the active channel, control of the generator terminal voltage shall transfer automatically to the inactive channel, as long as the inactive channel is considered available, without interruption of control, and this channel will now become the active channel. Upon restoration, the channel that assumed control shall remain in control of the generator terminal voltage unless a transfer is initiated by operator action or another failure or loss of power occurs. Normal mode enables automatic switching between channel A and channel B as needed. Regulation of the generator terminal voltage shall be maintained as specified in 3.3.3.9 during these transfers.

b. Single Channel – Channel A only. In "Single Channel – Channel A" mode, channel A is the active channel that shall monitor the generator terminal voltage and adjust the exciter field current to regulate generator terminal voltage.

c. Single Channel – Channel B only. In "Single Channel – Channel B" mode, channel B is the active channel that shall monitor the generator terminal voltage and adjust the exciter field current to regulate generator terminal voltage.

3.3.3.8 Cybersecurity. The voltage regulator system that utilizes IT equipment, as defined in DoDI 8500.01, shall have appropriate cybersecurity capabilities to counter threats (accidental or malicious) in accordance with the requirements specified below.

a. Changes to software, firmware, and programmable parameters shall require local authentication by the voltage regulator.

(1) Authentication shall be via a memorized secret using NIST SPECIAL PUBLICATION 800-63 or other means as approved by the purchaser. The memorized secret can be a password or PIN.

(2) Memorized secrets shall be changeable.

(3) The voltage regulator system shall have the ability to support memorized secrets of a minimum of eight characters.

(4) Factory default authentication credentials shall be removed or changed prior to delivery or installation.

b. All unused or unnecessary communication ports shall have the capability of being locked physically or disabled logically. A lockable external communication port shall be provided for maintenance functions. Access to all communication ports, except maintenance functions, shall be inside the locked enclosure.

c. There shall be no reprogramming capabilities over the connected supervisory network, remotely, or wirelessly.

d. Enclosures shall be capable of being locked physically using lead wire, tamper evident seals, hasps with padlocks, or other means.

e. When specified (see 6.2), the voltage regulator system shall have the capability to create, enable, modify disable, and remove user accounts. Controllers shall be able to support user accounts based on roles and responsibilities. Each account shall have rights in accordance with the principle of least privilege.

f. The voltage regulator system shall be configured in accordance with the most current version of the Defense Information Systems Agency issued Security Technical Implementation Guide and Security Requirements Guide.

g. The voltage regulator system shall be provided with authentication protocols for all communications with authorized devices.

h. Wireless networking capabilities shall be prohibited.

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i. Software and hardware that are not required for operation or maintenance shall be removed or disabled.

j. Voltage regulator system controllers shall be capable of validating the syntax and semantics input by users.

k. When specified (see 6.2), baseline configurations shall be provided. Baseline configurations of voltage regulator system controllers shall include, but not be limited to, the following: software packages, version numbers, patch information on operating systems and applications, configuration setting and parameters, network topology, and the logical placement of the components. All software shall include a cryptographic secure hash in accordance with FIPS PUB 180-4.

l. When specified (see 6.2), procedures to restore the voltage regulator system and its associated sensors shall be provided.

m. Voltage regulator systems shall have default parameters and programs stored in write-protected memory.

n. Unless otherwise specified (see 6.2), programmable parameters shall have an operating range that is modifiable. Exceeding the operating range shall generate an alert to the supervisory network, or locally for standalone voltage regulator systems.

o. Voltage regulator system controllers shall have on-board cybersecurity logging. The voltage regulator system controller shall be capable of supporting cybersecurity log retention for a minimum of 12 months unless otherwise specified (see 6.2). Cybersecurity logs shall be restricted to read only by users. Cybersecurity logged events shall include, at a minimum, any successful and unsuccessful authentications and successful and unsuccessful modifications. Cybersecurity event records shall include, at a minimum, the event time, event name, and username. Cybersecurity logs shall be exportable to removable media. Additional cybersecurity logging capabilities, if required, shall be as specified (see 6.2).

3.3.3.9 Operating mode transition. Automatic or operator-initiated operating mode transitions shall ensure the following:

a. Transitions between “Normal”, “Single Channel – Channel A”, and “Single Channel – Channel B” with the generator operating at any load from no load (or a minimal load sufficient to forward bias the main rectifier diodes as applicable) to rated load shall not result in transient voltages that exceed the limits set by 3.4.2 of this specification, damage any component, or trip the generator. A clear precedence of control between the two redundant channels of the voltage regulator system shall be ensured. Both channel A and channel B shall remain independent and continuously monitor the voltage regulator system parameters. Only the active channel can control the generator voltage and perform protective functions at any given time.

b. Transitions between “Normal”, “Single Channel – Channel A”, “Single Channel – Channel B”, or turning the voltage regulator system off with the generator operating at any load, from no load (or a minimal load sufficient to forward bias the main rectifier diodes, as applicable) to rated load, and at any speed, from 0 revolutions per minute (rpm) to rated rpm, shall not result in damage to any part of the generator or exceeding the overvoltage limit.

3.3.3.10 Operator controls and indications. At a minimum, the voltage regulator-exciter system shall provide the operator with the following operator controls and indications. Additional controls and indications proposed by the supplier(s) shall be subject to purchaser approval (see 6.2).

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3.3.3.10.1 Operating mode control switch. Control switches shall be in accordance with MIL-DTL-15291 for enclosure mounting and shall be rated in accordance with the table for class designation therein. Unless otherwise specified (see 6.2), a single or multiple control switch shall be provided to select regulator operating modes such as "Normal", "Single Channel – Channel A", and "Single Channel – Channel B". The switch contact arrangement shall be such that input signals to the switch for "Normal" enables automatic switching between channel A and channel B and that input signals for "Single Channel – Channel A" and "Single Channel – Channel B" operation can be electrically isolated. The design of the system and the switching arrangement shall be such that excitation control can be switched from any operating mode to another under any condition of load on the generator without causing personal injury or damage to any system component.

3.3.3.10.2 Field flashing. If applicable, for ship's service generator applications, field flashing power for initial voltage buildup shall be applied by means of a separate switch to electrically isolate the flashing source when not in use. Switches for applying field flashing power shall be in accordance with MIL-DTL-15291 for enclosure mounting. The switch shall be provided with either "Flash" and "Normal" or "Flash", "Normal", and "Off" positions. The "Flash" position shall be of the momentary type with forced spring return to the "Normal" position. Pushbuttons, if used, shall be of the normally open, double-pole type in accordance with MIL-DTL-2212 and suitable for back of board mounting and front of board operation. For emergency generator set applications, only devices that are completely static (no switches or relays) shall be used in the field flashing circuit to ensure that field flashing will be automatic and completely reliable, with the only exception being an "Off" position contact, to remove flashing current in the "Off" position. The voltage regulator shall be designed with a discrete input terminal internal to the regulator for rated speed indication and inhibit flashing without positive indication.

3.3.3.10.3 Voltage adjusting means. Separate manual adjustment of the generator voltage setpoint maintained by each channel of the voltage regulator system shall be furnished. The adjusting means provided shall be suitable for back of panel mounting and front of panel operation.

3.3.3.10.4 Manual backup voltage control. When specified (see 6.2), a manual voltage control, suitable for back of board mounting and front of board operation, shall be provided for control of the generator voltage without use of the basic voltage regulator and its associated components.

3.3.3.10.5 Direction of rotation – rotary control devices. When applicable, voltage adjusting devices shall be provided to increase the generator terminal voltage by turning to the right (clockwise) and decrease voltage by turning to the left (counterclockwise).

3.3.3.10.6 Control indications. The voltage regulator system shall be designed to provide indications as specified by the purchaser (see 6.2). At a minimum, the voltage regulator system shall include the following indications for each channel:

a. One or more indications of which voltage regulator (channel A or channel B) is the active channel in control.

b. An indication that a component of either channel A or channel B voltage regulator has failed and is unable to monitor the generator terminal voltage/control voltage.

c. One or more indications of control power availability.

d. An indication to show if the exciter is energized.

3.3.3.10.7 Remote operator controls and indications. The voltage regulator system shall be designed to support remote controls and indications as specified by the purchaser (see 6.2). At a minimum, the following controls and indications shall be provided. Additional remote controls and indications proposed by the supplier shall be subject to purchaser approval.

a. Raising and lowering the voltage regulator setpoint.

b. If provided, selecting between reactive differential compensation (see 6.4.g) and droop (see 6.4.d).

c. Commanding the voltage regulator to energize and de-energize the exciter field.

d. Voltage regulator status and faults.

3.3.3.10.8 Human-machine interface (HMI). When specified (see 6.2), controllers may utilize touchscreens or other forms or types of HMI. HMIs shall meet the requirements specified in MIL-DTL-32613.

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3.3.3.11 Endurance and heating. Unless otherwise specified (see 6.2), the voltage regulator shall meet 100-hour burn-in test requirements as specified (see 4.5.16). The maximum allowable temperatures shall be as specified in table I.

3.3.3.12 Insulation resistance. The insulation resistance, when tested in accordance with 4.5.18, shall be not less than 10 megohms.

3.3.3.13 Dielectric strength. Voltage regulator sets shall withstand the dielectric test voltage specified in 4.5.19. During and after completion of this test, there shall be no evidence of arcing, corona (visible or audible), or punctured insulation.

3.3.3.14 Shock. The voltage regulator-exciter system shall be able to withstand grade A, class 1, type A shock acceptance requirements of MIL-DTL-901 without change of operational state, mechanical or electrical damage, or electrical malfunction. Equipment shall not fail to function in any operational mode as a result of the shock test (see 4.5.20).

3.3.3.15 Vibration.

3.3.3.15.1 Type II system. The regulator and all switchboard mounted components shall be designed and tested to withstand externally excited vibration, type I, as specified in MIL-STD-167-1 (see 4.5.21). The rotating exciter shall be designed and tested for the same degree of vibration resistance as specified for the generator to which it is attached.

3.3.3.15.2 Type III system. The exciter, the regulator, and all switchboard mounted components shall be designed and tested to withstand externally excited vibration, type I, as specified in MIL-STD-167-1 (see 4.5.21).

3.3.3.16 Electromagnetic interference (EMI). To determine electromagnetic compatibility of the voltage regulator system with other shipboard components, the voltage regulator system shall comply with the EMI requirements of MIL-STD-461 when tested in accordance with 4.5.22.

3.3.3.17 Life. Wherever the expected life of parts is predictable and can be controlled through design of application, the parts shall be designed or applied so that equipment life of 40,000 hours of operation can be expected.

3.3.3.18 Configuration and excitation source.

3.3.3.18.1 Type II system. Unless otherwise specified (see 6.2), generator excitation power shall be obtained by rectifying the output of voltage and current transformers or combination transformers connected to the AC generator terminals. If specified, excitation power for turbine- and engine-driven generators may be furnished wholly from a shaft-mounted, fuse protected permanent magnet alternator. If specified, excitation power for a motor-driven generator may be furnished wholly from the motor power supply.

3.3.3.18.2 Type III system. Unless otherwise specified (see 6.2), generator excitation power shall be obtained by rectifying the output of voltage and current transformers or combination transformers connected to the AC generator terminals. If specified, excitation power for a motor-driven generator may be furnished wholly from the motor power supply.

3.3.3.18.3 Type III system flashing source. Unless otherwise specified (see 6.2), the system shall not depend on any voltage source external to the prime mover-generator set for flashing the AC generator in any mode of operation. Provision may be made for automatically flashing the AC generator field from the ship's service power supply during the starting of a motor-driven generator (see 6.2).

3.3.3.18.4 Control power. The voltage regulator system shall be designed to accept external independent redundant control power for all electronic circuits. In the event there is a loss of one of the control power sources, the other source shall be used to maintain voltage regulator operation using either control channel without interruption. A redundant control power source cannot be an internal uninterruptible power supply.

3.3.3.19 Voltage sensing. The voltage regulator shall sense the average of the three line-to-line voltages of the generator. The OVL circuit shall sense individual line-to-line voltages.

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3.3.3.20 Isolation. The voltage regulator, voltage adjusting circuitry, exciter (either type II system or type III system), and generator field shall be electrically isolated with independent power supplies by transformers from the generator armature output terminals. Control channels and redundant control power sources shall be electrically isolated with exception to any common signals that must be provided to both control channels.

3.3.3.21 Assembly. To facilitate troubleshooting and repair, the equipment shall, insofar as practicable, be packaged into functional assemblies. Such assemblies, except those that are plug-in assemblies, shall be wired to associated assemblies with screw terminal connections for ease of replacement.

3.3.3.22 Printed circuit board/printed wiring boards and assemblies. Unless otherwise specified (see 6.2), printed circuit boards, printed wiring boards, and associated assemblies shall be in accordance with MIL-DTL-917 with removable connection plugs. These connection plugs shall have latching mechanisms to ensure they cannot be removed unintentionally.

3.3.3.23 Multipin connectors. Unless otherwise specified (see 6.2), multipin connectors shall conform to SAE AS50151, MIL-DTL-26482, MIL-DTL-28748, MIL-DTL-55302, or MIL-DTL-83723, series 3. Connectors shall be selected such that the connector crimp or solder cups are of sufficient size to permit termination or the required wire size without cutting or otherwise deforming the wire to fit the crimp or cups.

3.3.3.24 Test and monitoring. As specified (see 6.2), the equipment shall include a test and monitoring capability to aid in monitoring circuitry for ease of maintenance. The equipment shall provide test points for control circuits for making measurements of voltage, current, and resistance as necessary for troubleshooting and adjusting operation of the equipment. Test points shall be provided at terminals arranged on a terminal board located in a readily accessible position. The test points shall be accessible to the user with any energized circuits adequately covered and safely isolated from the technician or less than or equal to 30 volts such that there is minimal risk to personnel. S9086-KC-STM-010/300 and MIL-HDBK-454 should be used as a guide.

3.3.3.25 Under-frequency operation. The regulator-exciter shall withstand without damage sustained operation at any frequency from rated frequency and below. Protective devices shall not be used to meet this requirement.

3.3.3.26 Over-frequency operation. The regulator-exciter shall withstand without damage sustained operation at frequencies from rated frequency to 125 percent of rated frequency. The regulator-exciter need not maintain the required steady-state voltage regulation (see 3.4.1) during overspeed operation above 105 percent of rated speed.

3.3.3.27 OVL. OVL shall be provided to limit the voltage in the event of a failure in the regulator or exciter. The voltage limiting feature shall be functional in any operating mode including “Normal”, “Single Channel – Channel A”, and “Single Channel – Channel B” modes of operation. The OVL system shall function independent of both the channel A and channel B voltage regulators.

3.3.3.28 Overexcitation protection (OEP). OEP shall be provided to isolate a generator if there is an overexcitation failure. The OEP feature shall be functional in any operating mode including “Normal”, “Single Channel – Channel A”, and “Single Channel – Channel B” modes of operation. The OEP system shall function independent of both the channel A and channel B voltage regulators.

3.4 Performance requirements.

3.4.1 Steady-state voltage regulation (see 6.4.h). Unless otherwise specified (see 6.2), the generator will operate anywhere from no load to rated kilowatt load at any power factor between 0.5 lagging and 0.8 leading. The average of the three line-to-line voltages at any loading shall be considered as the average voltage of the generator. The above limits apply for extended periods of time (minimum 8 hours) and for any mounting orientation of the regulator-exciter. When adjusted to operate at rated voltage, the average voltage maintained by the voltage regulator-exciter shall be within 3 percent of rated voltage for single-phase loading as specified in 4.5.5. The average voltage maintained by the voltage regulator-exciter after any or all blows of the class HI shock test shall be not more than 1 percent above or below the average voltage held before the shock test, without adjustment.

3.4.2 Transient voltage variation (dip or rise). With the regulator-exciter system in the “Normal”, “Single Channel – Channel A”, or “Single Channel – Channel B” mode, the maximum dip or rise in generator voltage for a given load change shall be in accordance with MIL-DTL-3124.

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3.4.3 Recovery time. The time to return to, and remain within, 3 percent of rated voltage following the application or removal of 2.0 per unit impedance load at 0 to 0.4 lagging power factor shall not exceed 1.5 seconds.

3.4.4 Short circuit current. The voltage regulator-exciter system shall be capable of maintaining, without damage to any part, three-phase and single-phase (line-to-line) short circuits at the generator terminals in accordance with MIL-DTL-3124 short circuit requirements. The system shall restore normal voltage upon removal of the short circuit.

3.4.5 Generator overload. The voltage regulator-exciter system shall maintain generator voltage within 5 percent of rated voltage in accordance with MIL-DTL-3124 overload requirements. Where a generator kilowatt overload rating is specified (see 6.2), the voltage regulator-exciter shall maintain rated generator voltage for the overload and duration specified.

3.4.6 Voltage adjustment range and steps. Unless otherwise specified (see 6.2), the voltage adjusting means furnished for the "Normal", "Single Channel – Channel A", and "Single Channel – Channel B" modes shall be capable of changing the voltage from 3.0 percent below to 7.0 percent above rated voltage in steps of not more than 1.0 percent of rated voltage at any load from no load to full load at rated power factor. This requirement shall be met with the reactive droop compensation, if provided, inoperative.

3.4.7 Parallel operation of generators. When parallel operation is specified (see 6.2), a reactive compensation function shall be provided, as specified (see 6.2).

3.4.7.1 Reactive droop compensation. When specified (see 6.2), an adjustable reactive droop compensation function shall be provided. The range of adjustment shall be from 0 to at least 10 percent of rated voltage in steps not to exceed 0.2 percent of rated voltage.

3.4.7.2 Reactive differential compensation. When specified (see 6.2), a reactive differential compensation function shall be provided.

3.4.7.3 Division of reactive (KVAR) load. The KVAR load of any generator (expressed as a percentage of its KVAR rating) shall not differ from the total KVAR load of all paralleled generators (expressed as a percentage of the total KVAR rating of all paralleled generators) by more than 5 percent as load is varied from 0 to 100 percent of the total KVAR rating of the generators at any power factor range as specified in 3.4.1.

3.4.8 Voltage waveform. The voltage regulator-exciter system shall contribute not more than 0.5 percent to the generator voltage deviation factor, nor more than 0.5 percent of any individual frequency voltage harmonic.

3.4.9 Power factor. The ship's loads are normally of lagging power factor. It is possible, however, that generators may occasionally operate at leading power factor. At rated kilowatt load with 0.8 leading power factor, the regulator-exciter shall be capable of maintaining generator terminal voltage within the steady-state voltage regulation band.

3.4.10 Protective function. The voltage regulator-excitation system shall provide the following protective functions for both channels across all operating modes in 3.3.3.7 regardless of which channel is the active channel controlling generator field voltage.

3.4.10.1 OVL. An OVL function shall be incorporated in the voltage regulator-exciter system. This function shall automatically limit generator voltage if a failure occurs in the voltage regulator-exciter system. The overvoltage limiter "ON" setpoint is evaluated as 105 to 107 percent of rated voltage. The overvoltage limiter "OFF" setpoint, or de-activation setpoint, is a programmable value set between 95 and 100 percent of the OVL activation setpoint. A pictorial representation of OVL setpoints is shown on figure 1.

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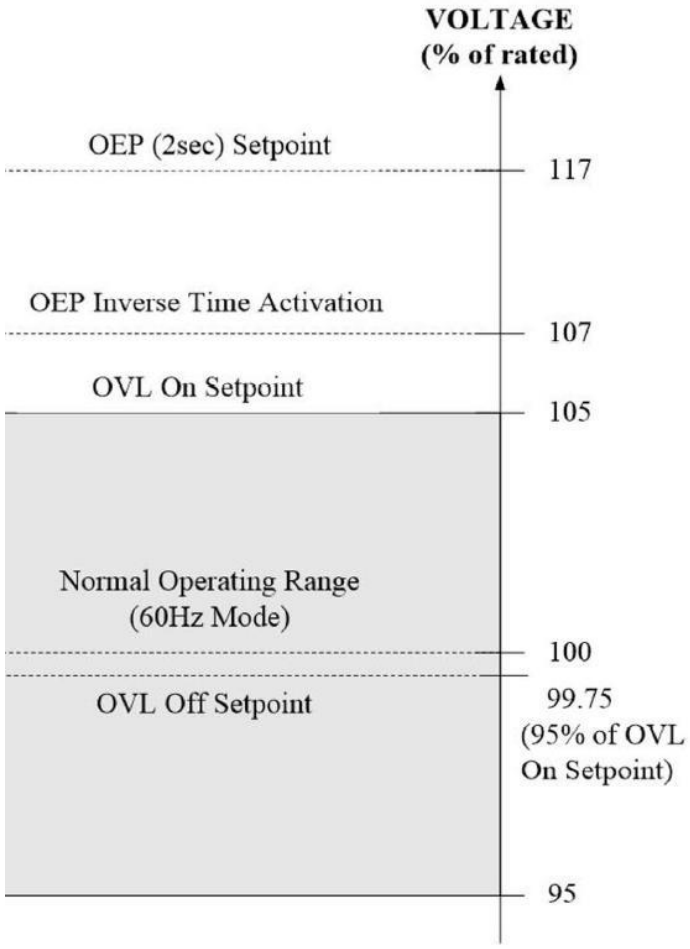


FIGURE 1. OVL and OEP setpoints.

3.4.10.2 OEP. When specified (see 6.2), the voltage regulator-exciter system shall provide an overvoltage protection feature to isolate a failed generator if there is an overexcitation failure. The OEP function shall use an inverse time delay function that is activated when the generator voltage reaches 107 percent of rated voltage regulator-exciter system voltage at the frequency of operation. The inverse time delay function integrates the accumulated overvoltage exponentially. An OEP timeout shall occur after 2 seconds of operation at 117 percent of the nominal voltage or after 6½ minutes of operation at 107 percent of rated voltage. A pictorial representation of OEP setpoints is shown on figure 1.

3.4.10.3 Undervoltage protection. When specified (see 6.2), the voltage regulator-exciter system shall provide an undervoltage protection feature to trip the generator output breaker and secure excitation in the event of a sustained undervoltage on the generator output. The voltage regulator-exciter shall be equipped with a generator undervoltage protection feature adjustable from 25 to 100 percent of rated voltage and with a time delay adjustable from 0 to 30 seconds. Unless otherwise specified (see 6.2), the nominal undervoltage protection shall be executed when rated voltage is below 80 percent for a period of 15 seconds or longer.

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3.4.10.4 Loss of sensing protection. When specified (see 6.2), the voltage regulator-exciter system shall be designed with the capability to monitor the state of all analog and digital input signals in order to identify the loss of one or more generator voltage sensors signals or generator current sensors signals and provide fault indication. Monitoring the input signals is useful in determining if a signal from an input device (such as a contact) is being properly received. Upon loss of all voltage or all current sensors, the voltage regulator-exciter system shall provide a discrete output terminal for generator output breaker tripping.

3.4.10.5 Exciter field overcurrent protection. When specified (see 6.2), the voltage regulator-exciter system shall include exciter field overcurrent protection.

3.4.10.6 Volts-per-hertz. When specified (see 6.2) the voltage regulator-exciter system shall include a volts-per-hertz limiter that takes control action to limit the volts-per-hertz ratio. The volts-per-hertz ratio limiter regulates voltage based on a purchaser-defined volts-per-hertz slope adjustable between 0.0 and 3.0 per unit.

3.4.10.7 Loss of excitation. In the event of loss of excitation, the voltage regulator-exciter system shall include a discrete output terminal for generator output breaker tripping.

3.4.10.8 Field flashing protection. When applicable, the voltage regulator shall terminate field flashing when the generator output voltage builds to a user programmable value of rated voltage. If the generator voltage does not build to the pre-set value within a field programmable time limit, the voltage regulator-exciter system shall provide an audible and visual indication of the condition and disable flashing.

3.4.10.9 Exciter rectifier diode fault detection. For type II exciters, the voltage regulator-exciter system shall incorporate a method for detecting a failed rectifier diode. Upon detection of diode failure, the voltage regulator-exciter system shall secure excitation and trip the generator output breaker.

3.4.11 Electric power system interface. The voltage regulator-exciter system shall meet type I, 60 hertz, three-phase electric power system interface test requirements of MIL-STD-1399-300-1. The electric power system interface tests required are grounding (susceptibility) test, user equipment power profile test, voltage and frequency maximum departure tolerance test, voltage and frequency transient tolerance and recovery (susceptibility) test, voltage spike (susceptibility) test, emergency conditions (susceptibility) test, current waveform (emission) test, voltage and frequency modulation (susceptibility) test, simulated human body impedance ground current test, and equipment line-to-ground voltage (susceptibility) test.

3.5 Identification plates. The identification plates shall be attached to a part of the equipment that will not ordinarily be renewed during its normal service life. These plates shall be located in a readily accessible position where they can be read at all times without danger to personnel. Identification plates shall be either type A, B, or C in accordance with MIL-DTL-15024 and MIL-P-15024/5. All engraved, stamped, or direct etched markings shall be filled with black paint, enamel, or lacquer. The information marked on the plates shall include the following items:

a. Manufacturer’s name, identification symbols, serial number, contract or order number, and date of manufacture.

b. Salient design characteristics; for example, type, frequency, voltage, and capacity.

c. Space for inspector’s official stamp.

d. National stock number, if available.

3.6 Workmanship.

3.6.1 General. Workmanship shall be in accordance with the requirements herein applicable to soldering, marking of parts and assemblies, wiring, welding and brazing, plating, riveting, finishes, machine operations, screw assemblies, and freedom of parts from burrs, sharp edges, or any other damage or defect that could make the part (or equipment) unsatisfactory for the purpose intended.

3.6.2 Threaded parts or devices. Screws, nuts, and bolts shall show no evidence of cross threading, mutilation, or detrimental or hazardous burrs.

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3.6.2.1 Tightness. Screw-type fasteners shall be tight. The word “tight” means the screw shall be firmly secured, and there shall be no relative movement possible between the attached parts.

3.6.3 Wiring. Insulated wire shall be formed into cables or ducted wherever practicable. Wires and cables shall be positioned or protected to avoid contact with rough or irregular surfaces.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on the first voltage regulator-exciter of a given design, type, and size when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests specified in table II. First article inspection shall be required after any change in design or use with a new generator design that affects the performance characteristics. First article inspections shall be repeated only for those performance characteristics affected by the change in design.

4.3 Conformance inspection. Conformance inspection shall be conducted on every voltage regulator-exciter equipment offered for delivery (see 3.3.1). Conformance inspection shall include the examination of 4.4 and the tests specified in table II.

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TABLE II. First article inspection and conformance inspection.

Examination and test	Requirement	Test method	First article inspection	Conformance inspection
Visual examination	3.2, 3.2.2, 3.3.2.1, 3.3.3.3 to 3.3.3.7	4.4	X	X
Creepage and clearance distances	3.3.3.2.5	4.5.1	X	--
Protective functions, OVL, and OEP	3.4.10, 3.3.3.27, 3.3.3.28, 3.4.10.1, 3.4.10.2	4.5.2	X 1/	X 2/
Operational modes (automatic and manual mode)	3.3.3.1.2, 3.3.3.7, 3.3.3.9	4.5.2	X 2/	X 2/
Cybersecurity	3.3.3.8	4.5.28	X 2/	X 2/
HMI	3.3.3.10.8	4.5.28	X 2/	X 2/
Steady-state voltage regulation	3.4.1	4.5.3	X 1/	X 2/
Frequency effect	3.4.1	4.5.4	X 1/	X 2/
Average of phase voltages	3.4.1	4.5.5	X 1/	--
Voltage waveform	3.4.8	4.5.6	X 1/	--
Voltage adjustment	3.4.6	4.5.7	X 1/	X 2/
Voltage transient and recovery time	3.4.2, 3.4.3	4.5.8	X 1/	--
Generator short circuit	3.4.4	4.5.9	X 1/	--
Leading power factor	3.4.9	4.5.10	X 1/	--
Under-frequency operation	3.3.3.24	4.5.11	X 1/	--
Over-frequency operation	3.3.3.26	4.5.12	X 1/	--
Parallel operation	3.4.7	4.5.13	X 1/	--
Generator overload	3.4.5	4.5.14	X 1/	X 2/
Reactive droop compensation	3.4.7.1	4.5.15	X 1/	--
Endurance and heating	3.3.3.11	4.5.16	X 2/	--
Enclosure	3.3.3.2.4	4.5.17	X	--
Insulation resistance	3.3.3.12	4.5.18	X	X
Dielectric strength	3.3.3.13	4.5.19	X	X
Shock	3.3.3.14	4.5.20	X	--
Vibration	3.3.3.15	4.5.21	X	--
EMI	3.3.3.16	4.5.22	X	--
Maintainability	3.3.2.4.3	4.5.23	X	--
Electric power system interface tests	3.4.11	4.5.25	X 1/	X 2/
Isolation	3.3.3.20	4.5.27	X 2	--

NOTES:

1/ These tests shall be conducted with the generator, or with the generator and prime mover, for which the regulator-exciter system is intended.

2/ These tests may be conducted with the generator, or with the generator and prime mover for which the regulator-exciter system is intended, or a simulated generator operation may be used (see 4.5.24).

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4.4 Visual examination. A careful examination shall be made of the materials, safety to operating personnel, and workmanship to ascertain that they are of the quality, and meet the applicable requirements, specified in this specification. The principal features of the design such as terminal connections, case construction mounting, nonstructural parts, assembly, and identification plate data shall be checked for compliance with this specification. These examinations shall not require disassembling the unit in such a manner that its performance, durability, or appearance would be affected.

4.5 Test procedures. Test procedures and test reports shall be provided as specified (see 6.2).

4.5.1 Creepage and clearance distances. Creepage and clearance distances shall be demonstrated by actual measurement to be in accordance with MIL-DTL-917 (see 3.3.3.2.5).

4.5.2 Protective functions, automatic transfer, manual mode transition. Protective functions, automatic transfer, and manual mode transition shall be verified as follows:

a. Protective functions. Verification testing of all protective functions (see 3.4.10) and communications (as applicable), including time delay tests, shall be conducted. These verification tests may be conducted with the generator and prime mover for which the voltage regulator is intended or a simulated generator and prime mover operation as specified in note 2 of table II.

b. Automatic transfer and manual mode transition. The voltage regulator automatic transfer function and manual mode transition shall be demonstrated to be compliant with this specification (see 3.3.3.1, 3.3.3.7, and 3.3.3.9). The automatic transfer function shall be demonstrated on each voltage regulator (channel A and channel B) by simulating a fault in the active voltage regulator. During this test, proper indications shall be verified including indication of which voltage regulator is active and expected fault indications. When volts-per-hertz mode is specified (see 6.2), the tests shall be performed in both 60 hertz and volts-per-hertz mode. The automatic transfer function test shall be conducted as follows:

(1) The generator used for the test shall be at rated speed (rpm) and no load. The voltage regulator shall be operating and regulating the generator terminal voltage at rated voltage with the voltage droop compensation function (if provided) operative and set at 4-percent droop.

(2) The generator speed, terminal voltage and current, load, and exciter field voltage and current shall be recorded prior to, during, and immediately after each simulated failure, loss of power, and each manual mode transition.

(3) A failure of a single channel of the voltage regulator (as determined by the supplier) to cause automatic transfer to the other channel shall be simulated. Following automatic transfer to the other channel of the voltage regulator, the channel that initially had the failure shall be restored with all faults cleared. The same test shall be performed with the other channel as the active channel initially; i.e., channel A as the active with a simulated failure for the first test, then channel B as the active channel with a simulated failure for the second test.

(4) De-energization of a single channel of the voltage regulator system to cause automatic transfer to the other channel. Following automatic transfer to the other channel of the voltage regulator, the channel that was de-energized shall have its power restored. The same test shall be performed with the other channel as the active channel initially – i.e., channel A as the active when de-energized, then channel B as the active channel with de-energized.

(5) Manual mode transitions shall be performed between the modes “Normal”, “Single Channel – Channel A”, “Single Channel – Channel B”, and “Off” using the operating mode control switch.

(6) The above tests shall be repeated at rated full load.

c. Control power. The voltage regulator control power redundancy shall be demonstrated to be compliant with this specification (see 3.3.3.18.4). For this test, control power momentary loss shall be tested to ensure the second, external redundant control power source maintains voltage regulator operations without interruption.

d. OVL. The OVL function shall be verified it performs as designed in both 60 hertz and volts-per-hertz mode. As specified (see 6.2), an analysis shall be made of the voltage regulator-exciter system to determine that no single failure will result in voltages in excess of the overvoltage limits specified in 3.4.10.1. To determine compliance with 3.4.10.1, the following tests shall be conducted:

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(1) With the generator operating at no load, rated voltage, and rated frequency, a failure in the voltage regulator to cause overvoltage shall be simulated. The overvoltage limit shall be set at the minimum and maximum limits and then approximately three equally spaced points to determine satisfactory operation.

(2) The following minimum readings shall be taken before, during, and after the simulated failure: the three line-to-line voltages, frequency, field current, and field voltage.

e. OEP. The OEP function shall be verified it performs as designed across frequency and voltage, in both 60 hertz and volts-per-hertz mode. As specified (see 6.2), an analysis shall be made of the voltage regulator-exciter system to determine that no single failure will result in voltages in excess of the overexcitation limits specified in 3.4.10.2. To determine compliance with 3.4.10.2, the following tests shall be conducted:

(1) With the generator operating at no load, rated voltage, and rated frequency, a failure in the voltage regulator to cause overvoltage shall be simulated. The overvoltage limit shall be set at the minimum and maximum limits and then approximately three equally spaced points to determine satisfactory operation.

(2) The following minimum readings shall be taken before, during, and after the simulated failure: the three line-to-line voltages, frequency, field current, and field voltage.

4.5.3 Steady-state voltage regulation. Steady-state voltage regulation tests shall be conducted as follows:

a. The generator used for the test shall be loaded with a second generator or a balanced impedance load at rated frequency and any convenient power factor.

b. The test shall be conducted with the rotating and control equipment at room temperature. The reactive droop compensation function, if provided, shall be inoperative.

c. The voltage regulator-exciter system shall be in the “Normal” mode with the channel A active and with no load on the generator. Adjust the average voltage and the frequency to their rated values; no further adjustments shall be made to the voltage regulator system during the remainder of the test. The first set of readings shall be taken within 2 minutes after the equipment is energized. The equipment shall then be operated in this no-load condition for 30 minutes after which another set of readings shall be taken.

d. Load shall be applied to the generator from no load to rated full load, rated field current, in five approximately equal steps, and back to no load in the same manner. A set of readings shall be taken at each point after voltage and current levels have stabilized. The average of the three voltages read at each point shall be considered the voltage for that load point.

e. The minimum readings to be taken during the tests shall be the three generator root mean square (rms) voltages and output line currents, frequency, kilowatt output, and generator field voltage and current (when available), to determine compliance with 3.4.1.

f. The test shall be repeated with channel B as the active channel after the average of phase voltages test (see 4.5.5) is completed.

4.5.4 Frequency effect. Immediately following the test specified in 4.5.3, without making any adjustments and with the generator operating at no load, the generator frequency shall be varied from 95 to 105 percent of rated frequency in six steps, approximately equally spaced. This test shall be repeated either at full-rated kilowatt load and rated power factor, or with the generator operating at a load corresponding to rated full load and rated field current at any convenient power factor. At each point, the three-line rms voltages shall be read and the average obtained after voltage levels have stabilized to determine compliance with 3.4.1. The minimum readings taken during the test shall be the three generator line-to-line rms voltages, the three line currents, frequency, kilowatt output, and the generator field current and field voltage.

4.5.5 Average of phase voltages. Immediately following the test for the frequency effect specified in 4.5.4, without making any adjustments and with the generator operating at rated rpm, no load, a single-phase resistance load of 15 percent of the generator rated full load current shall be applied, rated full load current being the current is rated kilowatt and rated power factor. The minimum readings to be taken during this test at each set point shall be three line-to-line rms voltages, three line currents, kilowatt frequency, field current, and field voltage to determine compliance with 3.4.1.

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4.5.6 Voltage waveform. The reactive droop compensation function, if provided, shall be inoperative. With the generator operating at rated rpm, kilowatt, and unity power factor, the voltage waveform and spectral analyses of the generator shall be taken. With the generator operating at rated rpm, kilowatt, and power factor and with the regulator-exciter disconnected and the generator field supplied by a direct current power supply, the voltage waveform and spectral analysis of the generator shall be recorded. The results of these two tests shall be compared to determine compliance with 3.4.8.

4.5.7 Voltage adjustment range and steps. A voltage adjustment range and steps test shall be conducted as follows:

a. The reactive droop compensation function, if provided, shall be inoperative. With the voltage regulator-exciter system in the "Normal" mode, with channel A active, and the generator at rated rpm at no load:

(1) Operate the voltage adjustment device at the rated voltage.

(2) Operate the voltage adjustment device from 3 percent below to 7 percent above rated voltage in 1-percent steps.

b. Repeat a(1) and a(2) for rated load conditions.

c. Repeat tests of 4.5.7 with the voltage regulator exciter system in the "Normal" mode with channel B as the active channel after step 4.5.7.1.

4.5.7.1 Minimum readings. The minimum readings taken at each step during the tests shall be the three generator line voltages, the three line currents, frequency, kilowatt, field voltage and field current, and division numbers on the voltage adjust rheostat (if used), if it is marked.

4.5.8 Voltage transient and recovery time.

4.5.8.1 General. The generator shall be operated with its intended voltage regulator-exciter system. The reactive droop compensation function, if provided, shall be inoperative.

4.5.8.2 Load application and removal. Load application and removal shall be conducted as follows:

a. With the generator at rated speed, voltage, and no load, a balanced three-phase load having an impedance not greater than 2 per unit and a power factor of 0 to 0.4 lagging shall be applied suddenly, to determine compliance with 3.4.2 and 3.4.3.

b. Repeat 4.5.8.2.a with sudden removal of the same load, without adjustment of the generator rpm or of the voltage.

c. If reactive droop compensation is provided, transient tests 4.5.8.2.a and 4.5.8.2.b shall be repeated with reactive droop compensation set at 4 percent.

4.5.8.3 Minimum readings. The minimum readings that shall be taken during each test are as follows:

a. Steady state. Before and after each load application or removal; three line-to-line generator rms voltages, three line currents, frequency, kilowatt, field voltage, and current shall be recorded.

b. Transient. At least one of the line-to-line voltages, all three line currents, frequency, a timing trace, and field voltage shall be determined and recorded. Data shall be recorded from at least 1 second before the instant of load application or removal to at least 4 seconds of operation after the instant of application or removal of load voltage.

4.5.8.4 Calculation. From oscillograms, five peak-to-peak measurements of voltage, at least five cycles apart, immediately preceding the instant of load change shall be made. These measurements shall be made by measuring the peak-to-peak displacement between two lines drawn through the center of the light spots of the peaks of the voltage and current waves that define the envelope of the trace. The average of these five readings will give the reference reading before the load change. Similar measurements shall be made immediately following the load change in the following manner: one measurement after one cycle, then five measurements spaced two cycles apart, and additional measurements spaced five cycles apart until steady state conditions within ±0.5 percent of the rated voltage exist. These readings divided by the reference reading shall be considered as per unit values during the transient. These per unit values versus time shall be plotted by a power monitoring device. A curve shall be drawn through these plotted points. A horizontal straight line shall be drawn-through the average of the results plotted for

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the last recorded second of readings, which should be for a time interval starting at least 3 seconds after the load was applied or removed in order to obtain the recorded oscillographic voltage or current for the final sustained reference. The limits of 3.4.2 and 3.4.3 shall also be drawn on this graph.

4.5.9 Generator short circuit test. The generator short circuit test shall be conducted as follows:

a. The generator at rated rpm, rated voltage, and no load shall be suddenly short circuited, three-phase. The short circuit shall be maintained for a period of time as determined by the formula in 3.4.4.

b. Repeat the test in 4.5.9.a, except that the generator shall be initially operating at rated load, unity power factor.

c. Repeat the test in 4.5.9.a, except that the short shall be applied line to line (rather than three-phase) and maintained for a period of time determined by the formula in 3.4.4.

d. The minimum readings to be taken during each test shall be as follows:

(1) Steady-state readings, before and after the short circuit; three line-to-line voltages, three-line currents, frequency, kilowatt, field currents, and field voltage. During the short, one of the line currents, field currents, and field voltage shall be taken.

(2) Transient. During the short, one of the line voltages, one of the shorted-line currents, field voltage, and field current shall be taken by means of a power monitoring device to record the data. The deflections of the oscillograph elements shall be as large as possible and consistent with obtaining legible oscillograms. The film speed shall be at least 0.5 foot per second. Sufficient film shall be used to record continuous operation from a least 0.5 second before the instant of short-circuit until the voltage has fully recovered and stabilized to ±0.5 percent of the rated voltage after the fault removal.

4.5.10 Leading power factor test. Leading power factor test shall be conducted as follows:

a. With the generator operating at rated speed and voltage, a leading power factor load shall be applied. The load shall be rated kilowatts and 0.8 power factor leading.

b. Minimum readings shall be taken at no load and at rated load: the three line-to-line voltages, the three line currents, kilowatt, frequency, field voltage, and field current.

4.5.11 Under-frequency operation. The generator, at no load and under the control of its intended voltage regulator-exciter, shall be operated at rated frequency to 0-hertz output in 5-hertz steps. The unit shall be operated at each point until temperatures stabilize to determine that the allowable temperatures of table I are not exceeded nor any other damage will occur. No other requirements apply to the under-frequency operation.

4.5.12 Over-frequency operation. The voltage regulator-exciter system shall be in the “Normal” mode. The generator shall be operated at no load in 2-hertz steps from rated frequency to that frequency corresponding to rated overspeed of the prime mover. The unit shall be operated at each point until temperatures stabilize to determine that the allowable temperatures of table I are not exceeded nor any other damage occurs.

4.5.13 Parallel operation. The parallel operation test shall be conducted as specified in MIL-G-3087 for steam turbine generators, MIL-G-21296 for diesel generators, MIL-G-22077 for gas turbine generators, or MIL-M-19097 for motor driven generators to determine compliance with 3.4.7.3 for reactive load division. The test shall be performed with the voltage regulator-exciter system of each generator in the “Normal” mode and the reactive compensation function of each system operative. The reactive compensation functions of all generators shall either be reactive droop compensation or reactive differential compensation, as specified (see 3.4.7, 3.4.7.1, 3.4.7.2, and 6.2). When volts-per-hertz mode is specified (see 6.2), the tests shall be performed in both 60-hertz and volts-per-hertz mode.

4.5.14 Generator overload. With the generator initially operating at rated voltage, rated frequency, no load, and with the voltage regulator-exciter in the “Normal” mode, generator overload shall be tested in accordance with MIL-DTL-3124 overload requirements at 0.5 to 0 power factor lagging to demonstrate compliance with 3.4.5. Readings of the three output voltages shall be taken prior to applying the load. At the end of the 2-minute test period, readings of the output voltages, line currents, field voltage, field current, and output power shall be taken prior to removing the test load.

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4.5.15 Reactive droop compensation test. With reactive droop compensation function operative, reactive droop shall be tested as follows:

a. At full load (0.8 lagging power factor), set the voltage regulator for rated voltage in the "Normal" mode.

b. Gradually reduce from rated load to no load in 10 percent steps.

c. Check and record rise in terminal voltage (voltage shall be recorded after initial transient).

d. Voltage rise shall be within the range of 2.7 to 4.0 percent of rated voltage.

4.5.16 Endurance and heating.

4.5.16.1 Endurance. The 100-hour burn-in test shall be conducted by energizing the voltage regulator and all electronic components of the exciter system (e.g., exciter rectifier circuitry, power rectifier circuitry, sensing circuitry, etc.) for 100 hours on each channel, totaling 200 hours. The endurance tests shall be made on each channel (channel A and channel B) of the voltage regulator-exciter system operated to control its intended generator continuously for 100 hours at the output terminal rated voltage and rated load without any adjustments. As an alternative, the test may be done as a separate component-level test that simulates operational electrical and thermal loads of the components. The unit shall be checked for any erratic operation, such as the inability to maintain the output terminal rated voltage and rated load without adjustments. A sufficient warm up period may be permitted prior to conducting this test to allow for any changes or drift in voltage regulator characteristics in going from cold to hot running conditions. If a failure occurs during this test, the equipment is to be repaired and the test continued from the point where the failure occurred.

4.5.16.2 Heating. Heating tests shall be made on the voltage regulator-exciter when operated at rated voltage and current conditions. At the supplier's option, the heating tests may be performed at the same time as the endurance test specified in 4.5.16.1. In those cases where the loading of certain circuit components may be greater at some low or intermediate load point than at rated load, additional tests shall be performed to demonstrate that these components are not overheating at these loads. The equipment shall be operated until the temperature has stabilized in all components under test. However, in no case shall the duration of the test be less than 4 hours. Temperature measurements shall be taken at 15-minute intervals. It shall be considered that stabilized temperatures have been reached when at least four consecutive readings taken at 15-minute intervals do not differ by more than 2 °C. Maximum temperatures taken shall be within the limits shown in table I when corrected to a 50 °C ambient.

4.5.16.3 Heat and endurance readings. The following minimum reading shall be taken during the heating tests. If the endurance test is run separately, then only letter a shall apply:

a. Three generator output voltages and line currents, frequency, kilowatt output, and generator field voltage and current.

b. Room ambient temperature.

c. Temperature of air entering and leaving any enclosure whose height exceeds 36 inches. The latter measurement shall be taken below the top cover and shall not exceed 4 inches from the top cover of the cubicle.

d. Thermometer or thermocouple readings. Where two or more identical components are used in a circuit (such as three single-phase transformers in a three-phase bank), readings shall be taken on at least ⅓ of the components or as recommended by the component manufacturer.

4.5.16.4 Measurement methods. Measurement methods used in the heating tests shall be as specified herein. For temperature measurements, detailed procedures of method 1 (thermometer) and method 2 (resistance) shall be as specified in MIL-DTL-917.

4.5.16.5 Ambient temperature. The ambient temperature shall be measured in accordance with MIL-DTL-917.

4.5.16.6 Semiconductor devices. The maximum permissible temperature (°C) shall be within table I limits, and the temperature rise of semiconductor devices shall be measured as follows:

a. The case or cell temperature required for semiconductor devices relying upon a cooling fin or other surface to dissipate heat from the case to the ambient shall be measured by placing a thermocouple in direct contact with the case.

b. Position on the case shall be on the high temperature side of the case.

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c. In no instance shall the thermocouple be placed in direct contact with the cooling fin or other heat dissipating surface.

4.5.17 Enclosure test. The enclosure test shall be as specified in MIL-STD-108 to determine if the enclosure is drip proof.

4.5.18 Insulation resistance. Prior to the application of the test voltage, all circuits shall be thoroughly discharged. The test voltage shall be applied between all electrically isolated circuits and between each circuit and frame (or chassis). Circuit diagrams shall be carefully studied prior to conducting this test to ascertain that circuits that may be isolated by transformers are not inadvertently neglected. When testing between the circuits and frame (or chassis), all circuits may be tied together so that only one test voltage needs to be applied, providing that the insulation resistance when tested in this manner meets the minimum value specified in 3.3.3.12. Insulation resistance shall be measured with an insulation resistance indicating instrument. The time of test voltage application shall be not less than 60 seconds. The temperature of the circuits at the time of the test shall be measured and recorded. Insulation resistance measurements shall be corrected to 25 °C. Corrections shall be made based on the insulation resistance doubling for each 10 °C decrease in temperature. The relative humidity at the time of the test shall be measured and recorded.

4.5.19 Dielectric strength test. The insulation resistance test of 4.5.18 shall be conducted before the dielectric strength test. The dielectric strength test shall be conducted on the completely assembled unit and not on an individual part. The test voltage shall be applied between electrically isolated circuits and between each circuit and frame (or chassis). The dielectric test voltage shall be twice the rated voltage plus 1,000 volts but not less than 1,500 volts applied for a period of 1 minute. The frequency of the test voltage and the wave form shall be not less than 60 hertz and shall approximate a true sine wave. Electrical components containing circuits rated 30 volts or less shall be exempt from dielectric strength testing.

4.5.20 Shock test. The shock test shall be conducted in accordance with MIL-DTL-901 and as follows:

a. General. The voltage regulator-exciter system or components thereof shall be energized during the shock test. The equipment may be tested when operating with its intended generator or under-simulated loading. All equipment or parts shall be mounted on the shock machine or platform in a manner simulating the actual installation onboard ship. The weight designation of the shock test shall be as required by the combined weight of the equipment or components thereof being tested. The output readings of the equipment shall be taken before, during, and after the test to determine compliance with 4.5.20.b.

b. Criteria for compliance with the shock tests. The voltage regulator-exciter system shall be considered as having met the shock test requirement unless any of the following failures are experienced. Any of the following failures shall be considered as cause for rejection of the equipment.

(1) Breakage of any parts, including mounting bolts. Cracking or signs of cracking in parts vital to operation shall be considered breakage.

(2) Appreciable distortion or dislocation of any parts such as mounting feet, coils, or conducting terminals.

(3) A value of insulation resistance (corrected to 25 °C) less than that permitted by 3.3.3.12.

(4) Failure to withstand a dielectric test voltage equal to 65 percent of that specified in 4.5.19.

(5) Failure to meet the steady-state voltage regulation requirements (see 3.4.1).

(6) Shift in the value of regulated voltage of more than 1 percent.

(7) Failure of the inactive channel's voltage control to meet the voltage adjustment requirements of 3.4.6 (see 4.5.7). If the inactive channel is required for voltage control, it shall also meet the steady-state voltage regulation requirements of 3.4.1 (see 4.5.3).

(8) Failure of any of the protective functions that may be provided to meet the requirements for OVL and under-frequency operation.

(9) Failure of the field flashing circuit, where provided.

c. Shock tests shall be performed at the contractor's plant, commercial laboratory, or Government laboratory that is equipped to perform these tests. When specified (see 6.2), copies of all master drawings shall accompany the equipment when shock tests are conducted at a location other than the contractor's plant.

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d. Correction of shock-tested equipment.

(1) Equipment that has been subjected to the HI shock test and has failed to perform any of the principal functions specified herein shall not be acceptable, either in whole or any of the parts, until it has been modified and successfully passed the HI shock test, or until the design modifications have been approved by the contracting activity.

(2) Equipment that has been subjected to the HI shock test and has successfully performed all the principal functions specified herein will be acceptable, provided the following post-shock tests are satisfactorily met:

(a) Dielectric.

(b) Insulation resistance.

(c) Steady-state voltage regulation. The steady-state voltage regulation test may be conducted with the test generator or by use of simulated loading conditions. When simulated means are used, an input-output form of test shall be conducted on the equipment prior to and again after the shock test to demonstrate that the electrical operating characteristics of the equipment meet the voltage regulation requirements of 3.4.1. To the extent possible, the equipment shall be energized and loaded to its rated values.

(d) Inactive channel voltage control.

(e) OVL.

(f) Under-frequency protection.

(g) Field flashing, where provided.

4.5.21 Vibration test. The voltage regulator-exciter equipment shall be tested in accordance with MIL-STD-167-1, type I, to demonstrate that the equipment meets the requirements of 3.3.3.15. The voltage regulator-exciter system or components thereof shall be energized during the vibration test. The equipment may be tested when operating with its intended generator or under-simulated loading. All equipment or parts shall be in a manner simulating the actual installation onboard ship. The basis of acceptability shall be in accordance with MIL-STD-167-1, type I, environmental vibration.

4.5.22 EMI characteristics tests. The voltage regulator-exciter system shall be subjected to CE101, CE102, CS101, CS114, CS116, RE101, RE102, RS101, and RS103 of MIL-STD-461. CS101 testing for the voltage regulator-exciter system shall include testing of voltage-sensing leads in addition to power leads because the voltage sensing path can be connected to the power distribution system via potential transformer (similar to power leads). Voltage regulator-exciter systems that have been tested to a previous or current revision of MIL-STD-461 and are capable of demonstrating compliance with the above requirements through prior testing need not be retested. Acceptance criteria: The voltage regulator-exciter system satisfies the above electromagnetic emissions and susceptibility requirements of MIL-STD-461.

4.5.23 Maintainability demonstration test. Maintainability shall be demonstrated (see 3.3.2.4.3), using MIL-HDBK-470 as a guide. A maintainability demonstration test report shall be submitted, if required (see 6.2).

4.5.24 Electric performance tests. Conformance tests may be conducted without the use of the generator (open-loop), rather than with a generator, as specified in note 2 of table II. These tests shall ensure that the requirements of the regulator-exciter system are met when operated with the intended generator. The following list is furnished for guidance to illustrate the nature of the tests that are to be conducted. Test procedures shall be provided, as specified (see 6.2). The composite test shall be of such nature to ensure that all polarity connections (primary and secondaries) and polarity of transformers, phase connection, and so forth within the equipment are correct.

a. Voltage regulator range and gain (input-output characteristic).

b. Static exciter range and gain (input-output characteristic).

c. Channel transfer voltage control operation.

d. Rectifier test.

e. Field flashing function check.

f. Feedback circuit polarity and time constant.

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g. Reactive load division function check.

4.5.25 Electric power system interface tests. The voltage regulator-exciter system shall be subjected to the electric power system interface test requirements of MIL-STD-1399-300-1, for type 1, 60-hertz, 115-volt, three-phase user equipment. If the voltage regulator-exciter system is protected with spike voltage limiters (varistors, etc.), the spike voltage test requirement shall be considered to be met if the voltage regulator system is not susceptible to values of peak voltage equal to the maximum safe level of the spike voltage limiter.

4.5.26 MTBF and reliability. The regulator-exciter shall be shown to meet the MTBF requirements for reliability as specified in 3.3.2.3. As noted in 3.3.2.3.1, the validation procedures in MIL-HDBK-217 may be used.

4.5.27 Isolation. An Insulation Resistance Test shall be performed in accordance with 4.5.18 and verify isolation of the dual, redundant channels and their associated circuitry. The channel not under test (channel A or channel B) shall be completely de-energized and isolated from the channel under test (channel A or channel B).

4.5.28 Cybersecurity and HMI. Cybersecurity and HMI requirements (see 3.3.3.8 and 3.3.3.10.8) shall be verified as follows:

(1) The ability to display the firmware version locally or via network interface shall be demonstrated, if applicable.

(2) The capability and operability of cybersecurity measures of 3.3.3.8 shall be demonstrated.

(3) For touchscreen HMIs, if installed, compliance with the requirements of 3.3.3.10.8 shall be demonstrated.

4.6 Toxicity. A health hazard assessment (HHA) will be conducted to ensure conformance to 3.2.3 as specified (see 6.2). The Navy and Marine Corps Force Health Protection Command (NMCFHPC) will evaluate the voltage regulator exciter system using data provided by the manufacturer/distributor to the NMCFHPC (see 3.2.3 and 6.9).

4.7 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.4 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Voltage regulator-exciter systems are intended to supply excitation power to, and control the output voltage of, ships' service generators on military ships and to control reactive load division when generators are operated in parallel.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Classification type required (see 1.2).

c. The specific issue of individual documents referenced.

d. If first article inspection is required (see 3.1 and 6.3).

e. Requirements for an HHA (see 3.2.3 and 4.6).

f. Requirement for prohibited materials verification (see 3.2.4 and 4.7).

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g. Documentation requirements (i.e., drawings, engineering and associated lists, first article inspection, test procedures and test reports, conformance inspection test procedures, and test reports, etc.) if required (see 3.3.2.3.1, 3.3.2.4.3, 3.3.2.5, 3.3.3.1, 3.3.3.1.1, 3.3.3.8, 4.5, 4.5.2.d, 4.5.2.e, 4.5.13, 4.5.20c, 4.5.23, 4.5.24, and 6.5).

h. Guidance for the accessibility of touchscreen digital interfaces (see 3.3.2.4.2).

i. FMECA requirements (see 3.3.2.5).

j. Type and arrangement of enclosure and mounting required (see 3.3.3.2.1 and 3.3.3.2.2).

k. If type of cooling is other than specified (see 3.3.3.2.5).

l. Cooling requirements (see 3.3.3.2.5).

m. When requirements for creepage and clearance distances are other than specified (see 3.3.3.2.6).

n. If protective devices, such as fuse protection, are required for generator terminal voltage sensing, line current sensing, and excitation circuit (see 3.3.3.6). The decision to use protective devices in these circuits should evaluate the tradeoffs between regulator-exciter system automatic protections and continued system operation.

o. When fuses are required to be other than in accordance with MIL-PRF-15160 (see 3.3.3.6).

p. When the capability to create, enable, modify disable, and remove user accounts is required (see 3.3.3.8.e).

q. When controller baseline configuration is required (see 3.3.3.8.k).

r. When controller restoration procedures are required (see 3.3.3.8.l).

s. When controller parameters range are other than specified, if applicable (see 3.3.3.8.n).

t. Controller logging capabilities, additional to those specified (see 3.3.3.8.o).

u. Additional requirements for operator controls and indications (see 3.3.3.10).

v. Operating mode control switch requirements (see 3.3.3.10.1).

w. When manual backup voltage control is required (see 3.3.3.10.4).

x. Control indications requirements (see 3.3.3.10.6).

y. Requirement for remote network (see 3.3.3.10.7).

z. HMI, if required (see 3.3.3.10.8).

aa. When the 100-hour burn in requirements are other than specified (see 3.3.3.11).

bb. Excitation source (see 3.3.3.18.1 and 3.3.3.18.2).

cc. Source of initial excitation if external to the generator set (see 3.3.3.18.3).

dd. Flashing source (see 3.3.3.18.3).

ee. When requirements for printed circuit boards, printed wiring boards, and associated assemblies are other than specified (see 3.3.3.22).

ff. Provide requirements for multipin connectors if other than specified (see 3.3.3.23).

gg. Provide requirements for test and monitoring capability for ease of maintenance (see 3.3.3.24).

hh. Provide generator power factor range if other than specified (see 3.4.1).

ii. Generator kilowatt overload rating (see 3.4.5).

jj. When requirements for voltage adjusting means are other than specified (see 3.4.6).

kk. Parallel operation required (see 3.4.7); type of reactive compensation required (see 3.4.7, 3.4.7.1, and 3.4.7.2).

ll. When OEP is required (see 3.4.10.2).

mm. When undervoltage protection is required (see 3.4.10.3); if nominal undervoltage protection is other than specified (see 3.4.10.3).

nn. When loss of sensing protection is required (see 3.4.10.4).

oo. When exciter overcurrent sensing protection is required (see 3.4.10.5).

pp. When volts-per-hertz is required (see 3.4.10.6).

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qq. If steady-state voltage regulation tolerance is other than specified (see 3.4.13.4.1).

rr. Requirement for providing testing procedures and reports (see 4.5 and 6.5).

ss. When voltage regulator-exciter system analysis is required (see 4.5.2.c and e).

tt. When volts-per-hertz mode is required (see 4.5.2.d and 4.5.13).

uu. Reactive compensation function requirements (see 4.5.13).

vv. Requirement for drawings to support shock test (see 4.5.20.c and 6.5).

ww. Packaging requirements (see 5.1, 6.6, and 6.8).

xx. Provisioning requirements, if required (see 6.6).

yy. Obsolescence and diminishing manufacturing sources and material shortages (DMSMS) analysis, if required (see 6.10).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from a specified quantity of first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.2. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract (see 6.2). Bidders should be told not to submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions. For the purpose of this specification, the following definitions should apply:

a. Active. A given channel of the voltage regulator-exciter system is said to be active if, and only if, it has been selected to control the generator field; only one channel can be active at any time.

b. Available. A given channel of the voltage regulator-exciter system is said to be available to control the generator field if, at a minimum, power is applied to that channel and no faults that prevent the controlling of the generator field are present on that channel.

c. Channel. One of two redundant halves of the voltage regulator-exciter system (channel A or channel B).

d. Droop. A means by which the voltage regulation setpoint is adjusted for generator KVAR loading. When droop is not zero, the voltage regulation setpoint is linearly lowered as generator loading is increased from zero to rated lagging power factor load and linearly increased as generator loading is increased from zero to rated leading power factor load.

e. Inactive. The inactive channel is not allowed to control the generator field. The inactive channel is determined by, when in single channel mode, the channel selector switch is set to the other channel. A channel is also inactive when, in normal mode the channel does not meet available criteria, or both channels meet available criteria and control has not transferred from the other channel.

f. Provisioning technical documentation (PTD). The data needed for the identification, selection, determination of initial requirements, and cataloging of support items to be acquired through the provisioning process. It includes provisioning lists and logistics support analysis summaries. Descriptive data such as drawings and photographs are referred to as "supplementary PTD".

g. Reactive differential compensation. Reactive differential compensation is a system that provides for the division of reactive load between paralleled generators by introducing a corrective effect, through the voltage sensitive element in proportion to the difference in the reactive load of each generator. A connection between each of the voltage regulator systems is required to provide the proper division of lagging and reactive load without introducing a drooping voltage characteristic.

h. Steady-state voltage regulation. Steady state voltage regulation is the change in sustained voltage expressed in percent of rated voltage, when the output of the generator is varied from no load to rated kW output at rated power factor and back to zero output with fixed settings for all adjustments of the voltage regulator. Voltage regulation is considered positive when the voltage increases with decrease in power output.

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i. Type II system – rotary brushless exciter and voltage regulator. A type II system is one that employs a rotary AC exciter, without brushes, and with rotating rectifiers. The regulator is composed of completely different static components.

j. Type III system – combined static exciter and voltage regulator. A type III system is one that supplies the generator excitation and controls the voltage through a completely static arrangement of transformers, reactors, and rectifiers, resistors, capacitors, transistors, and silicon-controlled rectifiers.

k. Unavailable. A given channel of the voltage regulator-excitation system is said to be unavailable to control the generator field if, at a minimum, no power is applied to the channel or faults that prevent the voltage regulator from controlling the generator field are present.

6.5 Recommended documentation. The following list of documentation should be considered for inclusion as part of the acquisition of any material under this specification to ensure supplies and services conform to this specification and only essential documentation/data is requested (see 6.2):

a. Reliability predictions report.

b. Maintainability predictions report.

c. Failure mode, effects, and criticality analysis report.

d. Drawings, engineering, and associated lists (level 3).

e. Test report (test/inspection).

f. Maintainability demonstration report.

g. Procedures (test/inspection).

h. Technical manuals (recommend military specifications and standards to be considered to ensure necessary content and format are provided).

i. PTD (see 6.4.f).

j. Obsolescence and DMSMS analysis.

6.6 Provisioning requirements. Acquisition documents should identify requirements for PTD, required quantity of spare parts, required quantity of repair parts, and support and test equipment required to operate and maintain an end item for an initial period of service (see 6.2). When ordering spare parts or repair parts for the equipment covered by this specification, such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment as specified (see 6.2). Packaging for such parts should also be as specified (see 6.2).

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 6.2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Conditions for use of level B preservation. When level B preservation is specified (see 6.2), this level of protection should be reserved for the acquisition of regulator-exciter systems for resupply worldwide under known favorable handling, transportation, and storage conditions.

6.9 Toxicity evaluation. The NMCFHPC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCFHPC to the Government for evaluation. The HHA process is described on the NMCFHPC's website, https://www.med.navy.mil/Navy-and-Marine-Corps-Force-Health-Protection-Command/Environmental-Health/Industrial-Hygiene/Acquisition-Technical-Support/Health-Hazard-Assessments-HHA/.

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6.10 Obsolescence. Re-procurement problems, closing of production lines, obsolescence of design, expected discontinuances of business by manufacturers, etc., in the operational environment cause problems in assuring sufficient supply of spare and repair parts. If these factors are determined to present potential problems, plans should be established early to assure that effective operational support will be available for the new system/equipment. MIL-HDBK-502, activity 14, establishes an obsolescence and DMSMS analysis process for identifying the loss, or impending loss, of manufacturers or suppliers of components, assemblies, sub-assemblies, piece parts and material required to operate and/or maintain the system/equipment. Establishment of a DMSMS program plan that proactively monitors candidate components and assemblies via engineering bill of materials and analyzes the best method to mitigate the loss of sources of supply, and creating a risk mitigation plan to counterfeit items. Establishment of a DMSMS program should be accomplished in conjunction with the requirements contained in TA-STD-0016. SD-22 should be considered for accomplishing MIL-HDBK-502, activity 14.

6.11 Subject term (key word) listing.

Control system, excitation

Control system, voltage

Excitation, brushless

Excitation, static

Regulator, voltage

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
(Project 6110-2022-003)

Review activity:
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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